"Calculus 3"

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 15

Any Reminders? Any Questions?

- Quiz on Friday on triple integrals in cartesian, cylindrical, and spherical coordinates.
- I will be away Wednesday-Thursday at a conference so no in-person class on 3/12 – watch videos instead!

ALVARO: Start the recording!

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Changes of Variables in Integrals

Today – Changes of Variables in Double and Triple Integrals

- Changes of Variables in Single Integrals
- Changes of Variables in Double Integrals
- Changes of Variables in Triple Integrals

Changes of Variables in Single Integrals (aka u-Substitution)

Example: Evaluate the (single) integral

$$\int_0^2 xe^{x^2}\,dx$$

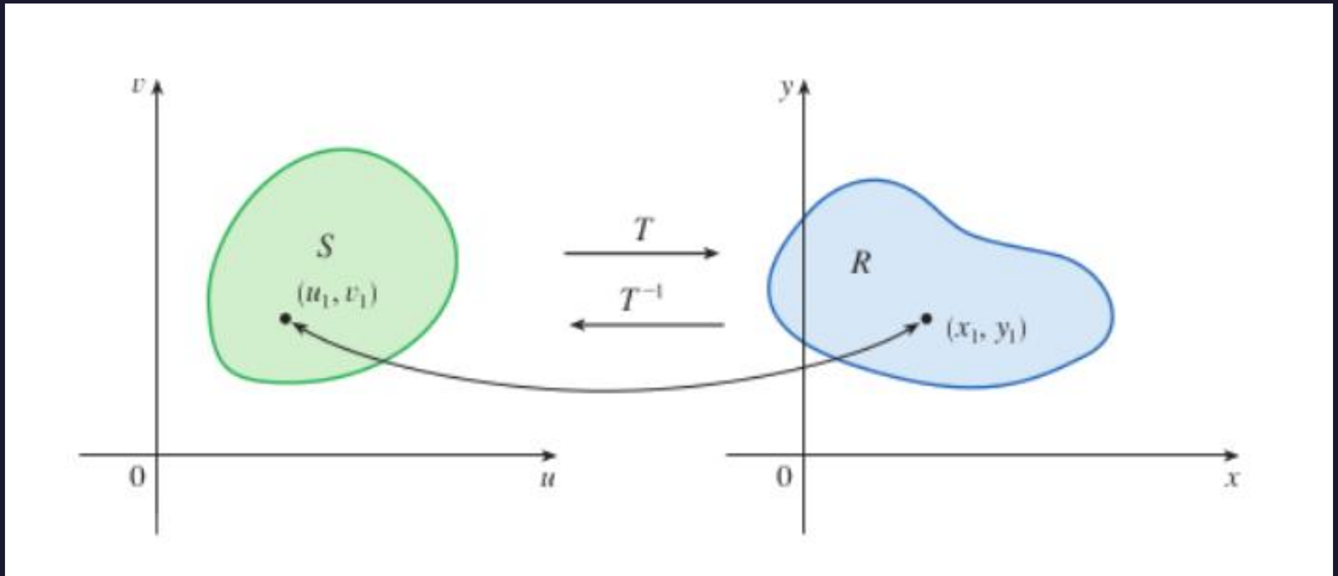
Changes of Variables in Double Integrals

Example: Evaluate the (double) integral

$$\int_{-1}^{1}\int_{0}^{\sqrt{1-x^2}} \sqrt{x^2+y^2}\,dydx$$

Changes of Variables in Double Integrals (in Polar)

Changes of Variables in Double Integrals (in General)



Example: Show that the change of variables $x = \frac{u}{2}, y = \frac{v}{2}$ sends the unit square (in x,y coordinates, in the first quadrant) to a square of side length 2 (in u,v).

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Example: Evaluate the (double) integrals

$$\int_0^1 \int_0^1 e^{2x+2y}\, dy\, dx \quad \text{and} \quad \int_0^2 \int_0^2 e^{u+v}\, dv\, du$$

Example: Use the change of variables $x + y = u, \quad y = v$ instead???

Example: Evaluate the (double) integrals

$$\int_0^1 \int_0^1 e^{2x+2y}\, dy\, dx \quad \text{and} \quad \int_0^1 \int_v^{v+1} e^{2u}\, du\, dv$$

Example: Show that the change of variables $x = R\cos(\theta)\,, y = R\sin(\theta)$ sends the square of side lengths R and pi/4 to a quarter circle of radius R. (Then, compute the areas of both regions.)

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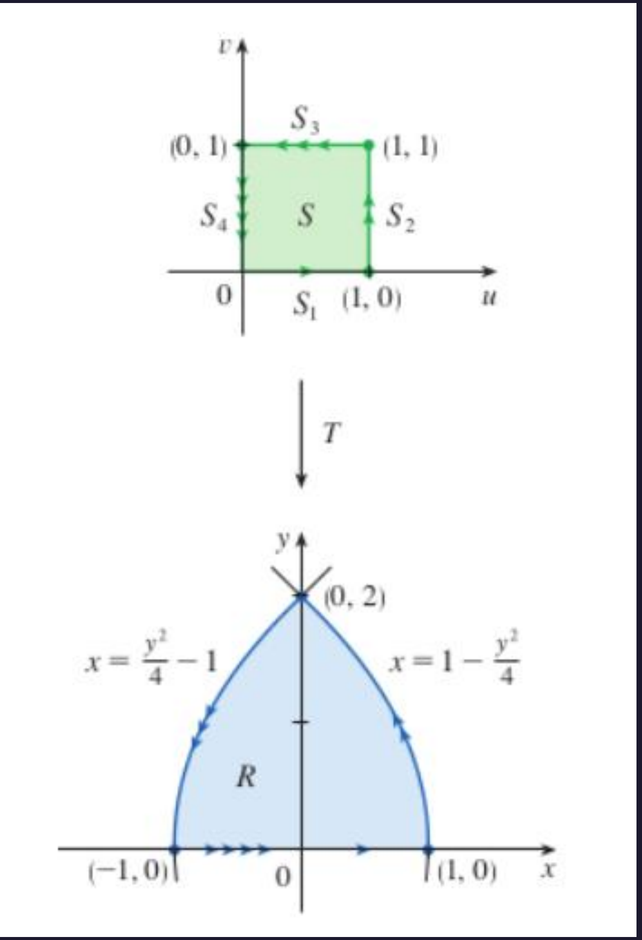
Example: Show that the change of variables $x = u^2 - v^2, y = 2uv$ sends the region S to the region R, and vice-versa.



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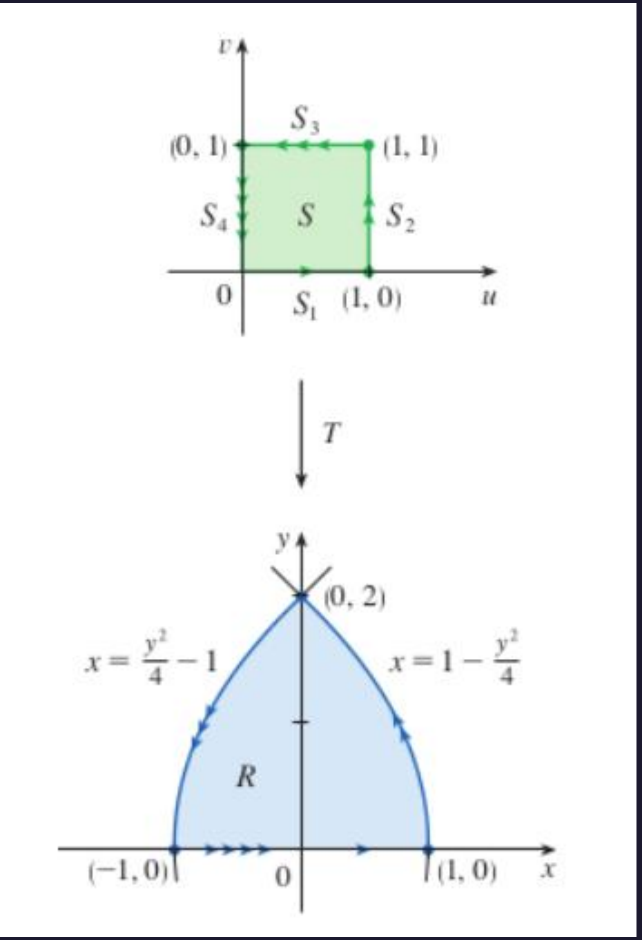
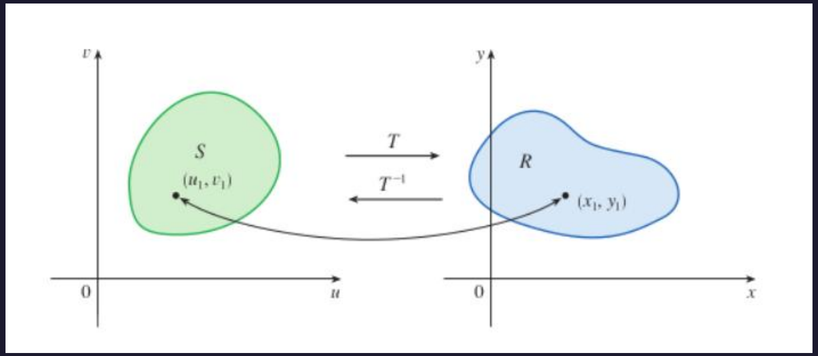


Changes of Variables in Double Integrals (in General)



Suppose that T is a C^1 transformation whose Jacobian is nonzero and that T maps a region S in the uv-plane onto a region R in the xy-plane. Suppose that f is continuous on R and that R and S are type I or type II plane regions. Suppose also that T is one-to-one, except perhaps on the boundary of S. Then

$$\iint_R f(x, y)\, dA = \iint_S f\left(x\left(u, v\right), y\left(u, v\right)\right) \left| \frac{\partial\left(x, y\right)}{\partial\left(u, v\right)} \right| du\, dv$$

Changes of Variables in Double Integral: the Jacobian

$$\iint_R f(x, y)\, dA = \iint_S f\left(x(u, v), y(u, v)\right) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du\, dv$$

The **Jacobian** of the transformation T given by $x = g(u, v)$ and $y = h(u, v)$ is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

The Jacobian is named after the German mathematician Carl Gustav Jacob Jacobi (1804–1851). Although the French mathematician Cauchy first used these special determinants involving partial derivatives, Jacobi developed them into a method for evaluating multiple integrals.

Example: Find the jacobian of the change of variables

$$x = R\cos(\theta)\,, y = R\sin(\theta)$$

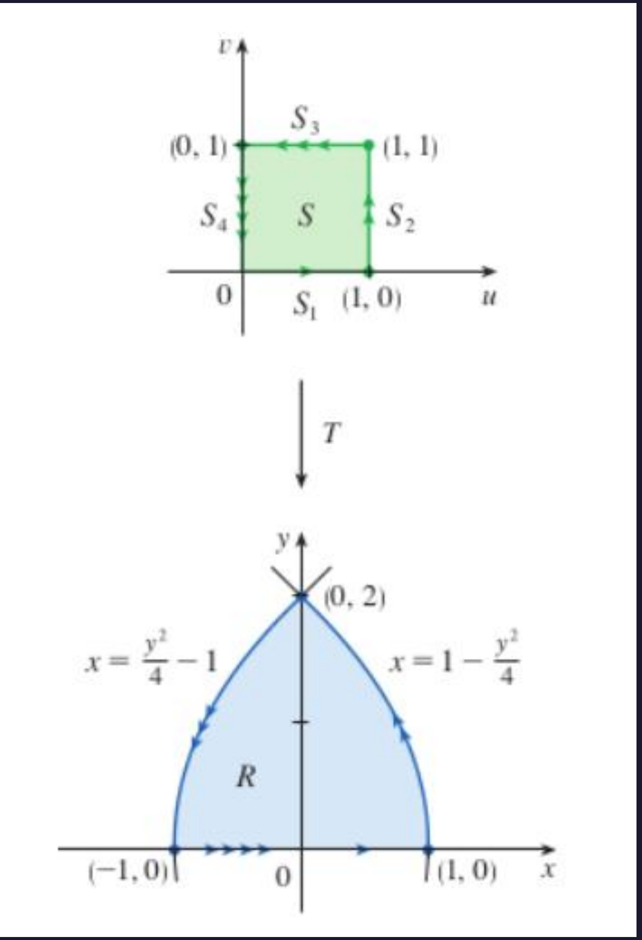
Example: Find the jacobian of the change of variables $x = \frac{u}{2}, y = \frac{v}{2}$

Example: Find the jacobian of the change of variables

$$x + y = u, y = v$$

Example: Find the jacobian of the change of variables

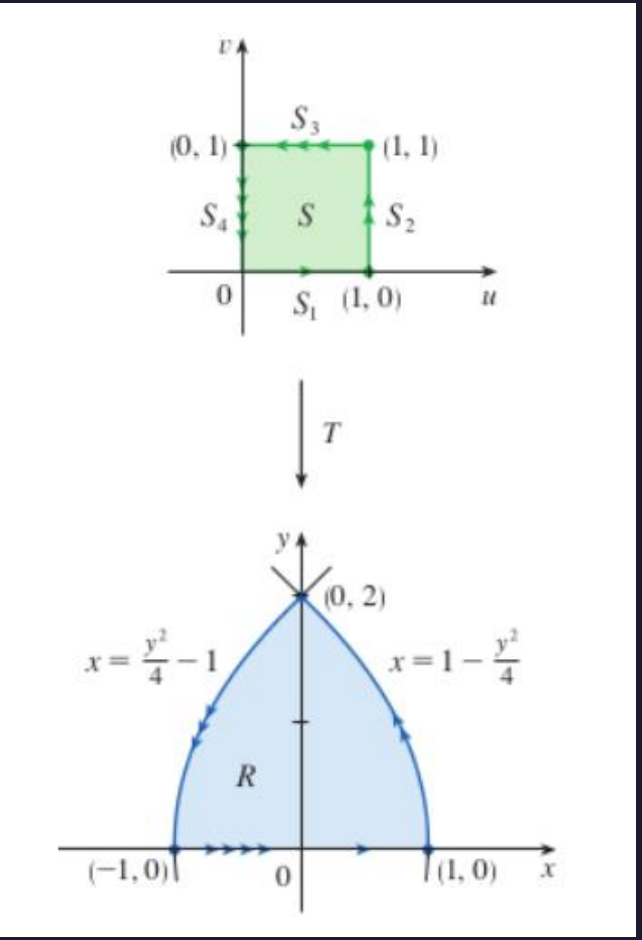
$$x = u^2 - v^2, y = 2uv$$



Example: Use the change of variables $x = u^2 - v^2, y = 2uv$ to evaluate

$$\iint_R y\,dA$$

where R is the region bounded by the parabolas $y^2 = 4 - 4x$ and $y^2 = 4 + 4x$ and $y \geq 0$.



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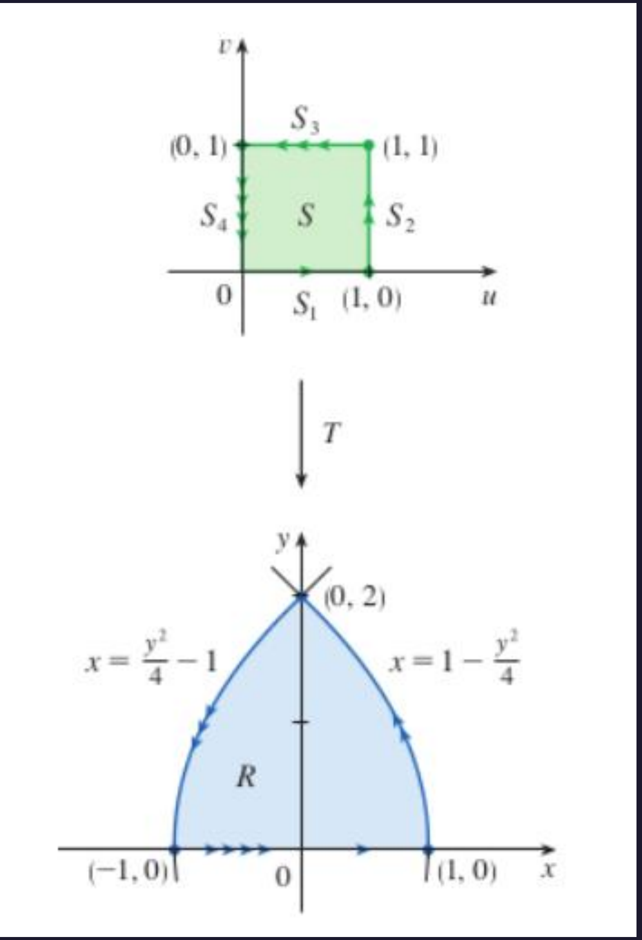

Example: Evaluate the integral $\iint_R e^{\frac{x+y}{x-y}}\, dA$ where R is the trapezoidal region with vertices $(1,0),\ (2,0),\ (0,-2),\ (0,-1)$.

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Example:

Use the change of variables $x = 2u, y = 3v$ to evaluate the integral

$$\iint_R x^2 \, dA,$$

where R is region bounded by ellipse $9x^2 + 4y^2 = 36$.

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where R is region bounded by ellipse $9x^2 + 4y^2 = 36$.

Example: Use the change of variables to evaluate the integral

$$\iint_R \sin(4x^2 + y^2)\,dA,$$

where R is the region in the first quadrant bounded by ellipse $4x^2 + y^2 = 1$.

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Changes of Variables in Triple Integrals

Changes of Variables in Triple Integrals (in General)

$$\iiint\limits_R f(x, y, z)\, dV = \iiint\limits_S f\left(x\left(u, v, w\right), y\left(u, v, w\right), z\left(u, v, w\right)\right) \left| \frac{\partial\left(x, y, z\right)}{\partial\left(u, v, w\right)} \right| du\, dv\, dw$$

$$x = g\left(u, v, w\right) \qquad y = h\left(u, v, w\right) \qquad z = k\left(u, v, w\right)$$

$$\frac{\partial\left(x, y, z\right)}{\partial\left(u, v, w\right)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

Example: Spherical Coordinates

$$x = \rho \sin \phi \cos \theta \qquad y = \rho \sin \phi \sin \theta \qquad z = \rho \cos \phi$$

We compute the Jacobian as follows:

$$\frac{\partial(x,y,z)}{\partial(\rho,\theta,\phi)} = \begin{vmatrix} \sin \phi \cos \theta & -\rho \sin \phi \sin \theta & \rho \cos \phi \cos \theta \\ \sin \phi \sin \theta & \rho \sin \phi \cos \theta & \rho \cos \phi \sin \theta \\ \cos \phi & 0 & -\rho \sin \phi \end{vmatrix}$$

$$= \cos \phi \begin{vmatrix} -\rho \sin \phi \sin \theta & \rho \cos \phi \cos \theta \\ \rho \sin \phi \cos \theta & \rho \cos \phi \sin \theta \end{vmatrix} - \rho \sin \phi \begin{vmatrix} \sin \phi \cos \theta & -\rho \sin \phi \sin \theta \\ \sin \phi \sin \theta & \rho \sin \phi \cos \theta \end{vmatrix}$$

$$= \cos \phi \left(-\rho^2 \sin\phi \cos\phi \sin^2\theta - \rho^2 \sin\phi \cos\phi \cos^2\theta\right)$$

$$-\rho \sin \phi \left(\rho \sin^2 \phi \cos^2\theta + \rho \sin^2 \phi \sin^2\theta\right)$$

$$= -\rho^2 \sin\phi \cos^2 \phi - \rho^2 \sin\phi \sin^2 \phi = -\rho^2 \sin \phi$$

Questions?

Thank you

Until next time.